

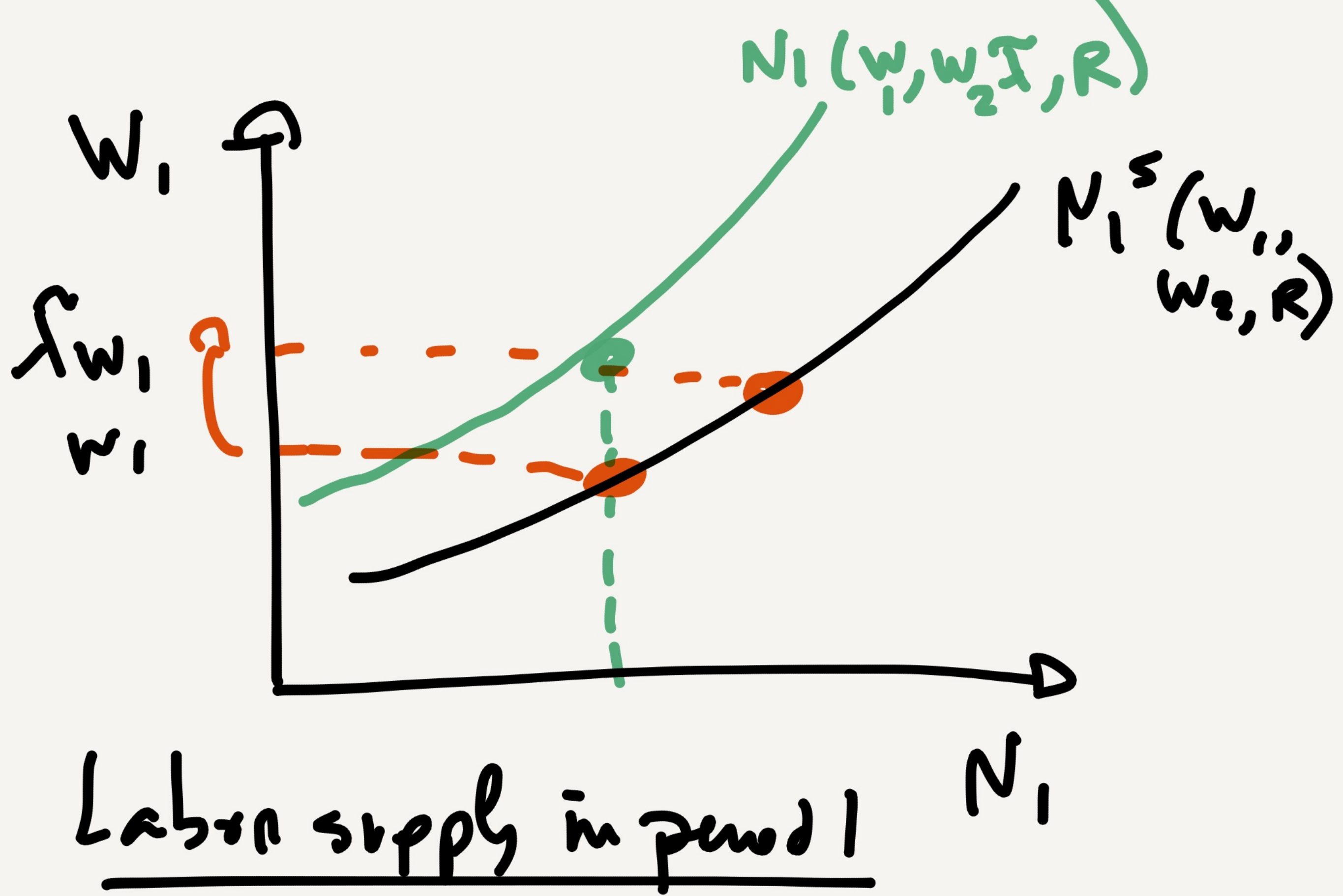
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M1 MACRO

Group A

Session 5

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.



Introducing a Gvt

$$p_1 G_1 + p_2 G_2 = p_1 T_1 + p_2 T_2 \quad \text{RB}$$

$$\left. \begin{array}{l} \end{array} \right\} G_1 = T_1 + B$$

$$\left. \begin{array}{l} \end{array} \right\} G_2 + (1+\tau_2)B = T_2$$

$$\hookrightarrow G_1 + \frac{G_2}{1+\tau_2} = T_1 + \frac{T_2}{1+\tau_2}$$

$$G_1 = T_1 + B$$

• $\nexists B > 0$, then $G_1 > T_1$

$\hookrightarrow G_1 - T_1 = \text{Budget deficit}$
 $B = \text{public debt}$

$$\underline{Mh} : P_1 C_1 + P_2 C_2 = \underbrace{P_1 (Y_1 - T_1)}_{\text{disposable}} + \underbrace{P_2 (Y_2 - T_2)}_{\text{income}}$$

$$U = \log C_1 + \beta \log C_2$$

$$\rightarrow \text{Max}_{C_1} \log C_1 + \beta \log \left[\frac{P_1}{2} (Y_1 - T_1) + Y_2 - T_2 - \frac{P_1}{P_2} C_1 \right]$$

$$\underline{\text{FOC}} \quad \frac{1}{C_1} = \beta \frac{P_1}{P_2} \frac{1}{C_2}$$

$$\Rightarrow C_2 = \beta \frac{P_1 C_1}{P_2}$$

↳ plug in the FOC of the HH:

$$P_1 C_1 + P_2 \left[\beta \frac{P_1 C_1}{P_2} \right] = P_1 (Y_1 - T_1) + P_2 (Y_2 - T_2)$$

$$C_1 = \frac{1}{1 + \beta} \left[(Y_1 - T_1) + \frac{P_2}{P_1} (Y_2 - T_2) \right]$$

and $C_2 = \dots$

Equilibrium

$$C_1 + G_1 = Y_1$$

$$\frac{1}{1+\beta} \left[Y_1 - T_1 + \frac{P_2}{P_1} (Y_2 - T_2) \right] + G_1 = Y_1$$

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$$\frac{1}{C_1} = \beta \frac{P_1}{P_2} \frac{1}{C_2}$$

$\Rightarrow$

Use eq:  $C_1 = Y_1 - G_1$   
 $C_2 = Y_2 - G_2$

$$\boxed{\frac{P_1}{P_2} = \frac{1}{\beta} \frac{Y_2 - G_2}{Y_1 - G_1}}$$